

a Argued trace

DeepSeek

Phi-4

Qwen3

Gemma-4

- clean twin: false trace installs
- ◆ true-fact twin: false trace installs
- implanted twin: true trace removes

0% 20% 40% 60% 80% 100%

share of the available room covered by an argued trace

b Irrelevant trace

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-10 -5 0 +5 +10

change in false-belief rate, irrelevant trace (pp)

